

Supplementary Material S1

Stochastic Morphological π -Mixture Kernels: Identifiability, Topological Stability, and Output-Space Learning

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Reproducibility guide | Strengthening branch Step 9 | Baseline archive v2.1.0 | Prepared 8 September 2026

Purpose and scope

This archive supplies the complete computational record accompanying the manuscript: frozen inputs, fitted models, candidate outputs, primary and diagnostic results, figure and table generators, and numerical checks of the mathematical claims. The written proofs remain in the main manuscript.

The original campaign evaluates a fixed morphological bank on a held-out test set. The subsequent output-space construction reuses that inspected set and is explicitly exploratory. Steps 5 and 6 add independent public-data validations: Oxford-IIIT Pet supplies higher-resolution human annotations under frozen synthetic corruptions, while DAVIS 2016 supplies genuine algorithm-produced segmentation errors with no synthetic corruption. Step 7 freezes the Step-5 IID-fitted selectors and evaluates nine structured Oxford error families without refitting or retuning. Step 8 then adds a validation-only fair-baseline audit against public SoftMorph2, an empirical 3×3 W-operator, and a condition-aware fixed-bank control. Step 9 closes the theory–computation loop with separately frozen finite stress tests of replication sensitivity, topology sharpness, incidence rank, Bayesian reversal, and kernel composition. Reproducing the original calculations does not create a new independent validation.

Primary experiment at a glance

Dataset	Mask size	Train / validation	Scored test
Shapes	48×48	1,000 / 300	499
MNIST	28×28	2,000 / 500	1,000
Fashion-MNIST	28×28	2,000 / 500	1,000

The 31-action bank is frozen before the final primary evaluation. Six primary corruption conditions produce 14,994 noisy cases; two additional conditions produce 4,998 diagnostic cases. Five fitted seeds (101-105) are retained for each dataset, together with three learned local-filter models.

Mean primary composite loss

Dataset	Validation-best fixed filter	Conditional selector
shapes	0.044996	0.040324
mnist	0.044702	0.042749
fashion	0.153080	0.130340

Lower is better. These are image-cluster means across primary corruptions and fitted seeds. The comparisons concern this fixed bank and study design, not all denoising methods. Full metric tables and paired intervals are in results/.

Protocol and indexing

Frozen data and exclusions

MNIST and Fashion-MNIST retain selected original pixels, thresholded masks (intensity ≥ 128), and original partition row IDs. Training and validation come from official training partitions; evaluation uses official test partitions. Development-pilot IDs and identical binary masks are quarantined. The shapes file retains 500 prospective test masks; ID 173 duplicates validation and is excluded from scoring, leaving 499.

Corruption schedule

Index / name	Deletion	Addition	Role
0 balanced04 / 1 balanced10	.04 / .10	.04 / .10	Primary
2 salt04 / 3 salt10	.01 / .01	.04 / .10	Primary
4 pepper04 / 5 pepper10	.04 / .10	.01 / .01	Primary
6 shift20 / 7 clean	.20 / 0	.20 / 0	Diagnostic

Training and validation use one corruption per image. Each final test image uses all eight conditions. The exact corruption law and seeded random-number construction are in `run_campaign.py`. Repeated corruptions of the same image are clustered for reporting.

Loss, topology, and uncertainty

The objective is $0.4(1-\text{Dice}) + 0.2(\text{binary MSE}) + 0.2$ times each capped, target-normalized component and hole error. Separately reported Betti-error columns contain raw absolute errors. Foreground uses 8-connectivity; background uses 4-connectivity with a permanent exterior frame. β_0 counts foreground components; β_1 counts bounded background components.

Sampled-method metrics are exact expectations over the finite categorical law. Mean-mask methods are scored separately. Aggregate PSNR is $-10 \log_{10}(\text{mean binary MSE})$. The bootstrap uses 2,000 clean-image cluster resamples; six paired control comparisons use Bonferroni-adjusted percentile intervals. These descriptive intervals do not certify performance on a new noise distribution.

Decode and join arrays correctly

Bank scores have axes `case × action × metric`. Metric order is loss, MSE, Dice, IoU, component error, hole error, exact topology. Action order is recorded in `environment.json`. Map `image_index` through the clean split IDs to recover a public source row. Primary CSVs use condition names; quotient CSVs use numeric condition indices.

Candidate masks are packed on the width axis in big bit order: unpack with `count=W`. Noisy masks are flattened before packing: unpack with `count= $H \times W$` and then reshape. Detailed field names and commands are in `docs/REPRODUCIBILITY.md`.

Reproduce the evidence

Environment and integrity

Use Python 3.12 and install the pinned requirements in a fresh virtual environment. No GPU or source-data download is needed for ordinary reproduction. The archive includes the clean masks and selected public pixels. The wrapper sets one thread for BLAS and OpenMP; fitted forests use `n_jobs=1`.

```
python -m pip install -r requirements.txt
python code/check_package.py
```

The checksum check verifies every frozen release file. Run it before making edits. The reproduction wrapper copies the package to a new directory and rejects an existing destination. All generated files and timings are written into that copy.

Three execution levels

```
python code/reproduce.py --level quick --workdir ../smm-quick
python code/reproduce.py --level analysis --workdir ../smm-analysis
python code/reproduce.py --level full --workdir ../smm-full
```

Level	Executed work
quick	Original mathematical checks; regenerated masks/features for two cases per dataset; all 15 forests replayed on those cases; frozen-pair invariance checked.
analysis	Quick checks, full saved-model replays, seed-101 fresh refits, all summaries/figures, expanded mathematical checks, quotient validation, and this guide.
full	Recompute all candidate banks from frozen clean masks, refit all models, then perform the analysis level.

Where each result comes from

`summarize.py` regenerates primary metrics, bootstrap intervals, ablations, diagnostic plots, qualitative examples, and gradient-variance diagnostics. `validate_kernel_quotient.py` regenerates the exploratory output-space table and saved probabilities. `verify_release.py` checks splits, reconstructed scores, all saved forest predictions, fresh seed-101 refits, and CPU timings.

`audit_math.py`, `audit_kernel_theory.py`, and `audit_oracle_risk.py` contain finite mathematical checks. `test_output_space_invariance.py` and `stress_test_output_space_invariance.py` verify the direct output-space representation invariance. `plot_topology_sharpness.py` generates the explicit extremal masks. `docs/RESULT_MAP.md` links evidence to exact filenames; `data/source_manifest.json` preserves source archive URLs and hashes.

Expected agreement

Discrete actions and masks should agree under the pinned stack. Saved float32 arrays introduce rounding: replay probability tolerances are 10^{-7} and sampled score recomputation tolerances are 2×10^{-6} . Timings, PDF metadata, and compressed-container bytes can differ. Compare scientific values rather than regeneration timestamps.

Kernel validation and limits

Exploratory output-space comparison

The output law uses the minimum predicted action risk within each exact-output class and a uniform reference over distinct outputs. Copying an existing frozen output/score pair preserves that law. Refitting after changing the feature representation is outside the invariance claim. Temperature is selected from .005, .02, and .1 using the original validation set; all 15 models select .005.

Dataset	Mean distinct outputs	Action Gibbs loss	Output-space loss
shapes	26.724	0.041460	0.041698
mnist	23.545	0.043322	0.043493
fashion	26.236	0.131275	0.131510

The output-space loss is slightly higher on all three datasets. Its demonstrated benefit here is invariance to redundant action representation. There are 14,880 direct replication checks on retained cases; the largest residual is below 7×10^{-16} . This extension is not an independent confirmatory benchmark.

Strengthening-branch direct learner and oracle checks

The table above remains the archived post-hoc class-minimum baseline. The strengthening branch additionally learns one scalar cost per distinct candidate-output row, so action duplication does not duplicate training observations. Under fixed deterministic fitting options/seeds, the manuscript proves end-to-end invariance under output-preserving action refinement. Step 3 verifies the construction across 57 representation forms, four mask sizes, three fitting seeds, duplicate multiplicities through 64 copies, 288 independent nonbase retraining comparisons, and 6,912 output-law comparisons; all recorded discrepancies are exactly zero at a required tolerance of 10^{-12} .

Step 4 adds a global oracle/excess-risk decomposition into finite-bank approximation, output-cost estimation, and entropy/reference terms. `audit_oracle_risk.py` checks the pointwise and global inequalities on 2,500 random finite problems containing 19,692 conditional inputs. No positive bound violation occurs; the exact-decomposition residual is 1.11×10^{-16} . These are mathematical/implementation checks, not additional independent benchmark data.

Independent Oxford and DAVIS validations

Step 5 uses all 3,669 official Oxford-IIIT Pet test masks at 128×128 under six frozen corruption conditions (22,014 held-out cases). The action-coordinate selector has mean composite loss 0.0162 versus 0.0164 for the validation-best fixed morphology; the direct-output selector is adverse at 0.0201 under both predeclared paired comparisons. Native/resized Betti pairs agree for 85.31% of Oxford test masks, so topology statements concern the resized targets.

Step 6 uses official DAVIS 2016 ground truth and pre-computed outputs from NLC, FST, SAL, TRC, MSG, and CVOS. No synthetic corruption is introduced. The held-out set contains 20 video sequences, 1,376 unique ground-truth frames, and 8,236 method-frame cases. The action-coordinate selector reduces sequence-mean composite loss from 0.3609 to 0.3071; the representation-invariant direct-output selector reaches 0.3104. Both improvements versus the raw segmentations have four-comparison family-wise 95% sequence-bootstrap intervals excluding zero. Region J and boundary F do not improve, while resized-target exact topology agreement rises from 0.1363 to 0.2339 for action-coordinate selection. The three adverse sequences are retained. Native/resized Betti pairs agree for only 26.24% of DAVIS held-out frames, so DAVIS topology metrics are explicitly tied to the 128×128 evaluation targets.

Step 7 structured-error distribution shift

Step 7 reuses the frozen Step 5 IID deletion/addition fitting distribution and fixed hyperparameters, but evaluates all 3,669 held-out Oxford masks under nine predeclared structured errors: boundary expansion/contraction, correlated false positives/negatives, interior holes, fragmentation cuts, a thin bridge/spur, localized occlusion, and a mixed error. No Step 7 structured case enters fitting or tuning.



The displayed mask is selected by a rule frozen before outcome analysis and is illustrative only. Full condition definitions are in docs/STRUCTURED_ERROR_STEP7.md.

Structured error	Input	Fixed	Action	Direct
Boundary expand	0.0591	0.0509	0.0512	0.0508
Boundary contract	0.0508	0.0501	0.0494	0.0546
Clustered FP	0.2458	0.2325	0.2337	0.1704
Clustered FN	0.2384	0.2119	0.2204	0.1613
Interior holes	0.1990	0.1898	0.1907	0.1577
Fragmentation cut	0.1180	0.1265	0.1262	0.0206
Bridge/spur	0.0098	0.0135	0.0117	0.0184
Occlusion	0.0850	0.0892	0.0889	0.0711
Mixed structured	0.4224	0.3953	0.4124	0.2046

Secondary natural component-dropout diagnostic ($n = 116$ eligible masks): input loss 0.0858, validation-best fixed 0.1054, action-coordinate 0.1038, and direct-output 0.1111. All topology-exact rates are zero because the diagnostic deliberately removes a genuine component; this diagnostic is excluded from the nine-condition headline mean.

The nine-condition mean favors the frozen direct-output selector: loss 0.1010 versus 0.1587 for input, while exact topology agreement rises from 0.3663 to 0.5287 and boundary F from 0.8759 to 0.9173. The family-wise direct-minus-input loss interval is $[-0.05888, -0.05650]$. Direct-output selection is adverse for boundary contraction and bridge/spur errors, and the secondary component-dropout diagnostic is also adverse; all are retained.

Step 8 modern and fair baselines

Step 8 uses the same Step 5 fitting/validation information budget and never tunes on the Step 7 structured errors. Public SoftMorph2 is retrieved at runtime; an independent product-logic implementation matches the public 2D code with maximum absolute discrepancy 0.0. Validation selects SoftMorph closing with 4-connectivity, one iteration, $\sigma = .75$, and threshold .5. A regularized empirical 3×3 W-operator is also selected on validation only, together with an intentionally advantaged IID condition-aware fixed-bank control.

On the 22,014 IID cases, condition-aware fixed morphology has loss 0.01574, action-coordinate selection 0.01619, direct-output selection 0.02006, SoftMorph2 0.09752, and the empirical W-operator 0.15709. Both learned selectors significantly outperform SoftMorph2 and the W-operator under the predeclared family-wise comparisons. On the 33,021 structured-OOD cases, direct-output selection remains at 0.10104 versus 0.15195 for SoftMorph2 and 0.16448 for the W-operator; direct-output significantly outperforms both. The adverse direction is retained: SoftMorph2 significantly outperforms action-coordinate selection under structured shift (0.15195 versus 0.15385).

Step 9 theorem-targeted finite stress tests

Step 9 is not a performance benchmark. It enumerates complete small state spaces where feasible and targets exact consequences of the paper's theorems. Replication tests use duplicate counts through 256 plus 4,500 random partitioned categorical laws; the sharp TV constant is attained to 1.1×10^{-16} while the distinct-output count remains fixed. Topology

checks enumerate all 2,304 Hamming-one pairs on 3×3 , add 30,000 random multi-flip cases, and recover both $|\Delta\beta_k| = 3$ sharpness constructions.

On all 512 3×3 masks, the 31 action labels induce 18 distinct maps and the incidence rank is 18; exact deduplication yields a full-rank 18-map bank, while adding a duplicate label leaves the rank unchanged. A four-action lossy kernel on the same 512-state space recovers the uniform prior exactly under Bayesian reversal but has $HH_\mu^\leftarrow$ residual 0.9953. Exact finite optimal transport on all 16 2×2 masks gives $\kappa(K) = 0.6$ and $\kappa(K^n) = 0.6^n$ through $n = 8$ for $K = .6I + .4O_{5 \times 5}$, while a deterministic 2×2 square opening has $\kappa = 4$. These checks illustrate both the contracting and noncontracting regimes; they do not assert global contraction of the trained 128×128 kernel.

Numerical support for the proofs

Independent graph calculations compare 66,146 masks and 1,053,698 single-pixel changes, check 2,304 exact local component identities, and include separate constructions attaining both sharp sensitivity bounds. The suite also runs 2,000 entropy/risk/replication trials, 32 finite transport comparisons, Bayesian-reversal checks, and gradient comparisons.

The largest finite-difference gradient residual in the expanded suite is 8.2×10^{-11} . These finite and numerical checks complement the quantified written proofs; they are not a proof-assistant certificate or external mathematical review.

Limits to retain in any presentation

Individual metrics do not all improve: MNIST Dice and shapes hole error can worsen. Stronger Fashion-MNIST corruption gives a documented failure against the fixed area filter. Clean inputs can change. Evaluating a full 31-action bank is materially slower than using a single area filter. Oxford closes the low-resolution-only gap, DAVIS closes the synthetic-error-only gap, Step 7 probes structured mask-error shift, Step 8 adds a matched public SoftMorph2/W-operator comparison, and Step 9 stress-tests theorem consequences on small finite state spaces; none establishes RGB restoration, clinical performance, correlated real-sensor-noise robustness, or native-resolution topology preservation. DAVIS source segmentations are from the 2016 benchmark rather than contemporary architectures.

Step 8 evaluates the public SoftMorph2 operator as matched post-processing and a learned local W-operator, but it does not reproduce full BiMoNN/DMNN architectures or SoftMorph as an end-to-end network component. No universal architecture-level superiority is claimed. The mathematical contraction statements require their stated coefficient conditions; the fitted model has not been shown to satisfy a global contraction coefficient below one.

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